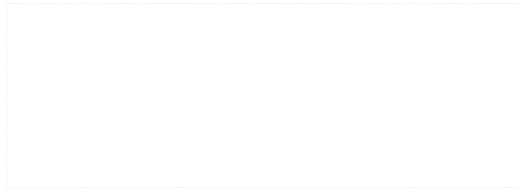


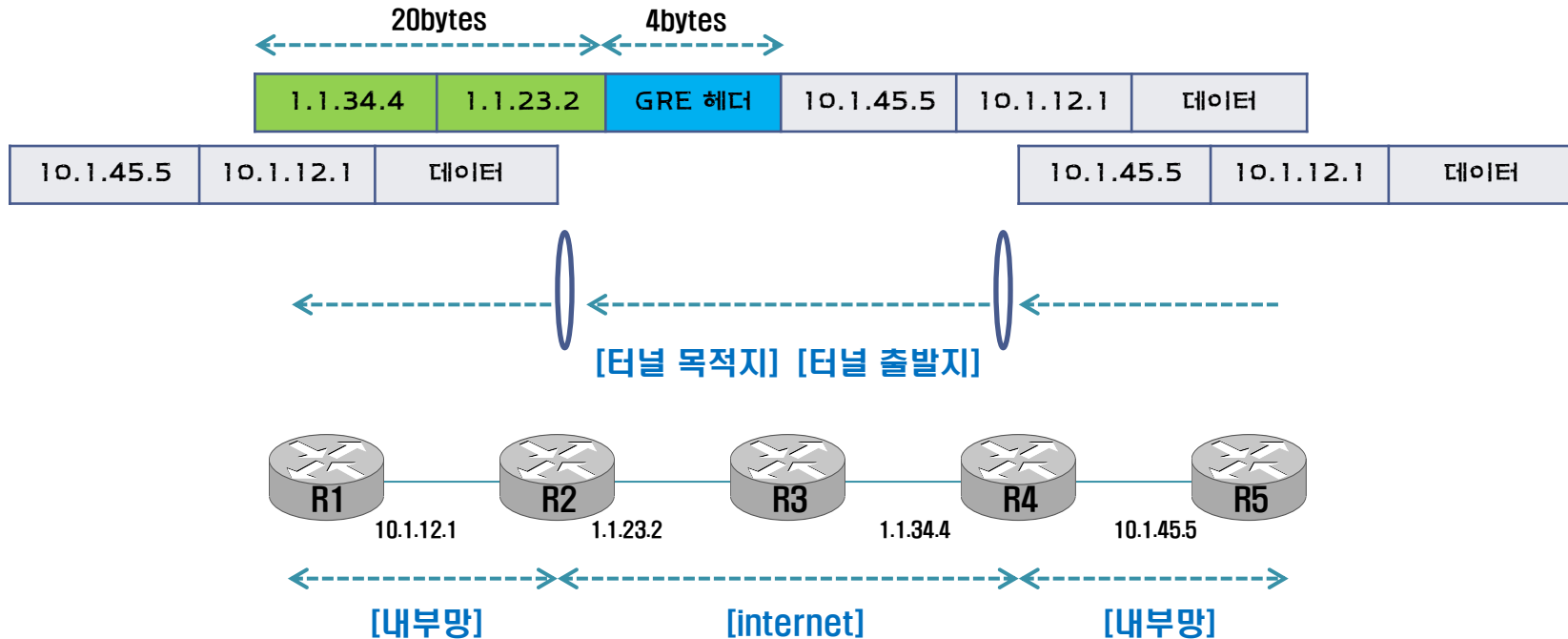


GRE IPSEC VPN



• GRE IPSec VPN이란?

: IP, IPv6, IPX, IPsec 등과 같은 여러 가지 패킷을 IP에 실어 전송할 때 사용하는 터널링 프로토콜



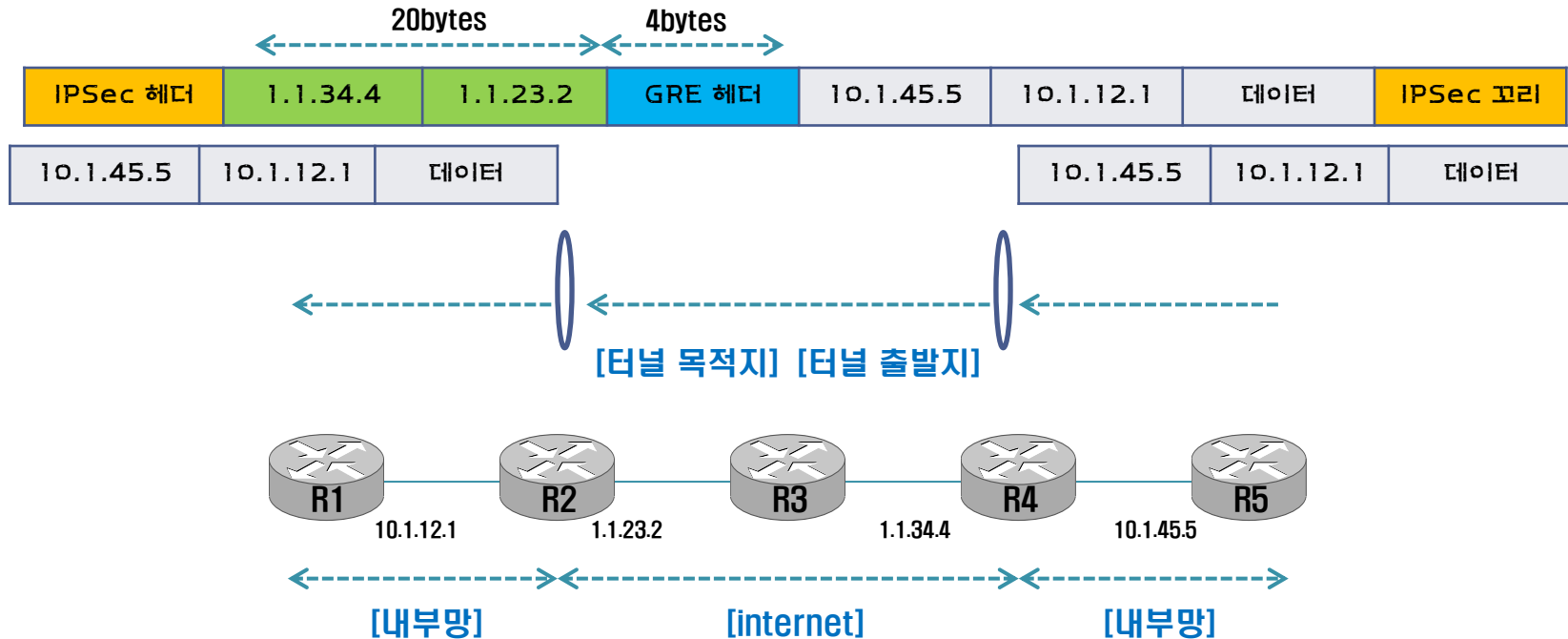
: GRE 패킷을 전송하는 도중의 라우터들은 터널의 목적지 IP 주소인 1.1.23.2를 참조하여 패킷을 라우팅

→ GRE 헤더 내부의 패킷이 어떤 주소를 사용하든 GRE를 위하여 추가된 라우팅 가능한 터널 목적지 주소만을 참조하므로 인터넷을 통하여 전송이 가능

: GRE 헤더에는 GRE 내부의 패킷 정보가 표시

• GRE IPsec 동작방식

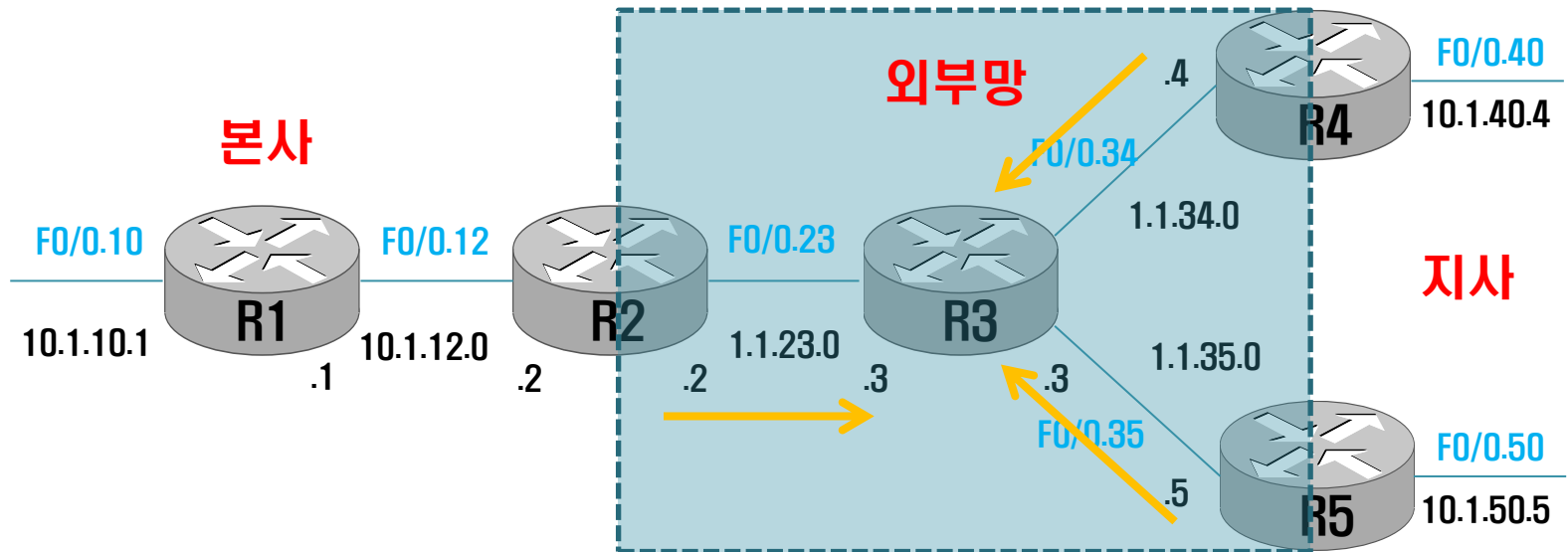
: GRE IPsec은 GRE 패킷을 다시 IPsec으로 보호



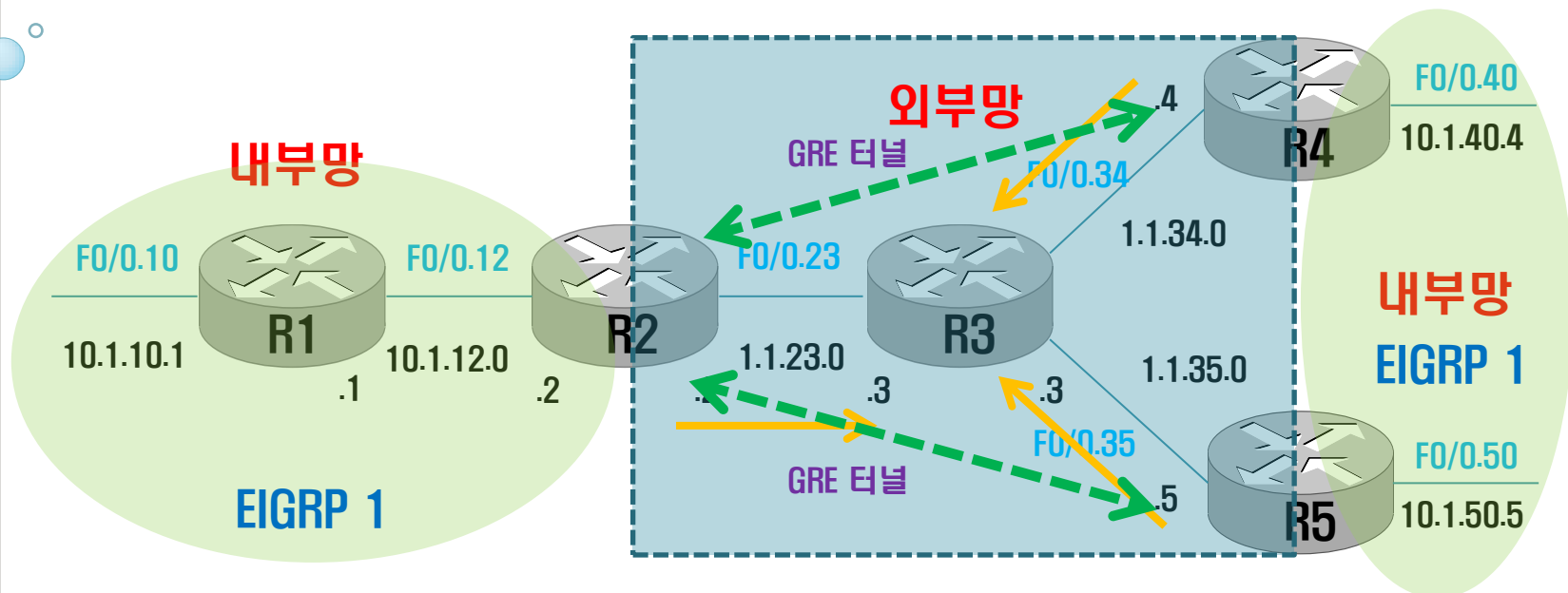
• GRE IPsec의 장점

- 동적인 라우팅 프로토콜 사용 가능
- 멀티캐스트 패킷 전송 가능
- 동적인 라우팅 프로토콜을 이용하여 이중화 구성 용이

• 테스트 네트워크 구축



• GRE 터널 구성



- 1) GRE 터널 설정모드로 들어감
- 2) 라우팅 가능한 IP 주소를 사용하여 터널의 출발지 IP주소 지정
- 3) 라우팅 가능한 IP 주소를 사용하여 터널의 목적지 IP주소 지정
- 4) GRE keepalive 기능 사용(옵션)

• GRE 터널 구성

//GRE터널 구성

<R2>

```
interface tunnel 24
tunnel source 1.1.23.2
tunnel destination 1.1.34.4
ip address 10.1.24.2 255.255.255.0
keepalive 10 3
!
```

```
interface tunnel 25
tunnel source 1.1.23.2
tunnel destination 1.1.35.5
ip address 10.1.25.2 255.255.255.0
keepalive 10 3
!
```

<R4>

```
interface tunnel 24
tunnel source 1.1.34.4
tunnel destination 1.1.23.2
ip address 10.1.24.4 255.255.255.0
keepalive 10 3
!
```

<R5>

```
interface tunnel 25
tunnel source 1.1.35.5
tunnel destination 1.1.23.2
ip address 10.1.25.5 255.255.255.0
keepalive 10 3
!
```

//확인

R2#ping 10.1.24.4

R2#ping 10.1.25.5

R2#show ip route

(생략)

1.0.0.0/24 is subnetted, 1 subnets

C 1.1.23.0 is directly connected,
FastEthernet0/0.23

10.0.0.0/24 is subnetted, 3 subnets

C 10.1.12.0 is directly connected,
FastEthernet0/0.12

C 10.1.25.0 is directly connected, Tunnel25

C 10.1.24.0 is directly connected, Tunnel24

S* 0.0.0.0/0 [1/0] via 1.1.23.3

• GRE를 통한 라우팅 설정

//GRE를 통한 라우팅 설정

<R1>

```
router eigrp 1
network 10.1.10.1 0.0.0.0
network 10.1.12.1 0.0.0.0
!
```

<R2>

```
router eigrp 1
network 10.1.12.2 0.0.0.0
network 10.1.24.2 0.0.0.0
network 10.1.25.2 0.0.0.0
redistribute static
!
```

<R4>

```
router eigrp 1
network 10.1.24.4 0.0.0.0
network 10.1.40.4 0.0.0.0
!
```

<R5>

```
router eigrp 1
network 10.1.25.5 0.0.0.0
network 10.1.50.5 0.0.0.0
!
```

//확인

R1#show ip route

(생략)

10.0.0.0/24 is subnetted, 6 subnets

C 10.1.10.0 is directly connected,
FastEthernet0/0.10

C 10.1.12.0 is directly connected,
FastEthernet0/0.12

D 10.1.25.0 [90/297246976] via 10.1.12.2,
00:00:23, FastEthernet0/0.12

D 10.1.24.0 [90/297246976] via 10.1.12.2,
00:00:23, FastEthernet0/0.12

D 10.1.40.0 [90/297249536] via 10.1.12.2,
00:00:16, FastEthernet0/0.12

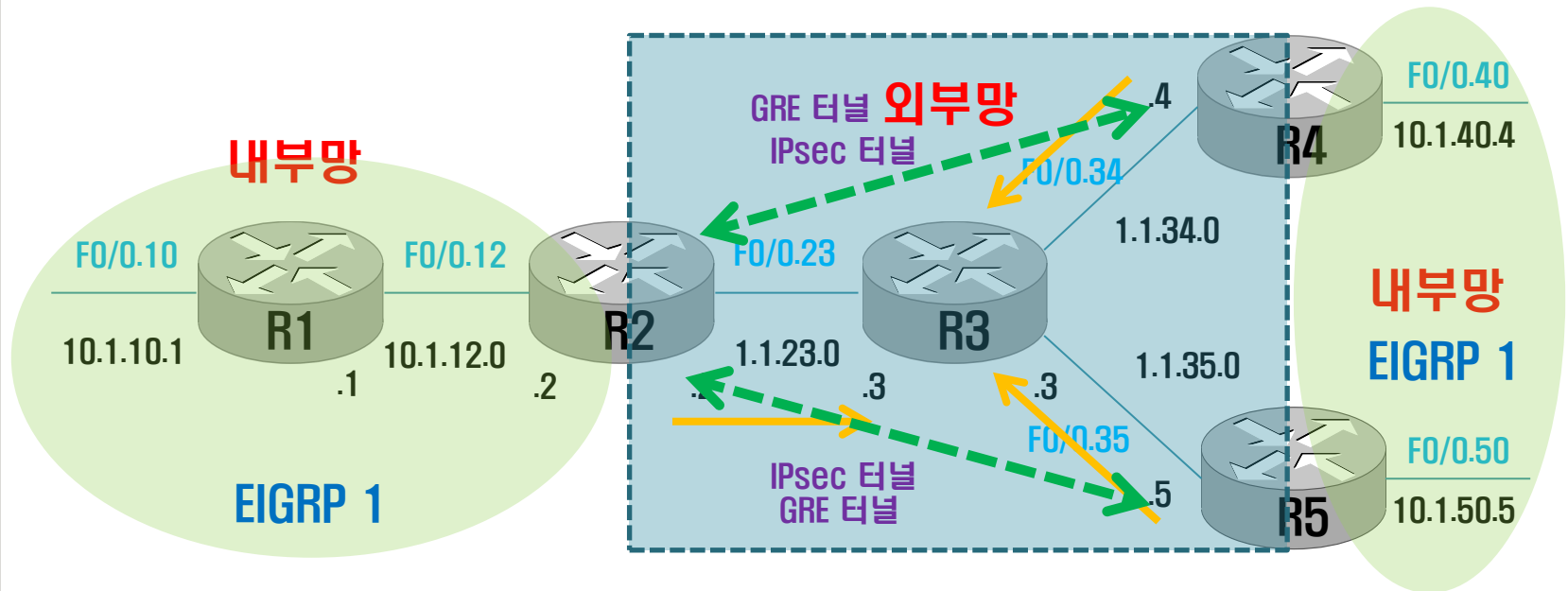
D 10.1.50.0 [90/297249536] via 10.1.12.2,
00:00:07, FastEthernet0/0.12

D*EX 0.0.0.0/0 [170/30720] via 10.1.12.2,
00:00:23, FastEthernet0/0.12

R1#ping 10.1.40.4

R1#ping 10.1.50.5

• GRE IPsec VPN 구성



- GRE IPsec VPN 구성

- 1) ISAKMP 정책 설정
- 2) 피어별 암호 지정
- 3) DPD(dead peer detection) 설정
- 4) IPsec으로 보호할 트래픽 지정
- 5) IPsec SA 지정
- 6) 정적 암호 맵 생성
- 7) 인터페이스에 정적 암호 맵 적용

• GRE IPsec VPN 구성

//GRE IPsec VPN 구성

<R2>

```
crypto isakmp policy 10
encryption aes
authentication pre-share
!
crypto isakmp key cisco address 1.1.34.4
crypto isakmp key cisco address 1.1.35.5
!
crypto isakmp keepalive 10
!
ip access-list extended acl_R4
permit gre host 1.1.23.2 host 1.1.34.4
!
ip access-list extended acl_R5
permit gre host 1.1.23.2 host 1.1.35.5
!
crypto ipsec transform-set trset_GRE esp-aes
esp-md5-hmac
!
```

```
crypto map cmap_GRE 10 ipsec-isakmp
match address acl_R4
set peer 1.1.34.4
set transform-set trset_GRE
!
crypto map cmap_GRE 20 ipsec-isakmp
match address acl_R5
set peer 1.1.35.5
set transform-set trset_GRE
!
interface f0/0.23
crypto map cmap_GRE
!
```

• GRE IPsec VPN 구성

<R4>

```
crypto isakmp policy 10
encryption aes
authentication pre-share
!
crypto isakmp key cisco address 1.1.23.2
!
crypto isakmp keepalive 10
!
ip access-list extended acl_GRE
permit gre host 1.1.34.4 host 1.1.23.2
!
crypto ipsec transform-set trset_GRE esp-aes
esp-md5-hmac
!
crypto map cmap_GRE 10 ipsec-isakmp
match address acl_GRE
set peer 1.1.23.2
set transform-set trset_GRE
!
interface f0/0.34
crypto map cmap_GRE
!
```

<R5>

```
crypto isakmp policy 10
encryption aes
authentication pre-share
!
crypto isakmp key cisco address 1.1.23.2
!
crypto isakmp keepalive 10
!
ip access-list extended acl_GRE
permit gre host 1.1.35.5 host 1.1.23.2
!
crypto ipsec transform-set trset_GRE esp-aes
esp-md5-hmac
!
crypto map cmap_GRE 10 ipsec-isakmp
match address acl_GRE
set peer 1.1.23.2
set transform-set trset_GRE
!
interface f0/0.35
crypto map cmap_GRE
!
```

• GRE IPsec VPN 구성

//확인

R2#show crypto isakmp sa

IPv4 Crypto ISAKMP SA

dst	src	state	conn-id	slot	status
1.1.34.4	1.1.23.2	QM_IDLE	1003	0	ACTIVE
1.1.35.5	1.1.23.2	QM_IDLE	1004	0	ACTIVE

R5#show ip ro eigrp

10.0.0.0/24 is subnetted, 6 subnets

D 10.1.10.0 [90/297249536] via 10.1.25.2, 00:00:25, Tunnel25
D 10.1.12.0 [90/297246976] via 10.1.25.2, 00:00:25, Tunnel25
D 10.1.24.0 [90/310044416] via 10.1.25.2, 00:00:25, Tunnel25
D 10.1.40.0 [90/310046976] via 10.1.25.2, 00:00:25, Tunnel25

//본사-지사는 포인트 투 포인트 GRE 터널을 사용하므로 지사간 통신은 본사를 통해 이루어짐

R5#ping 10.1.40.4

//GRE 패킷은 ESP에 인캡슐레이션 되어 전송되므로 매칭되는 게 보이지 않음

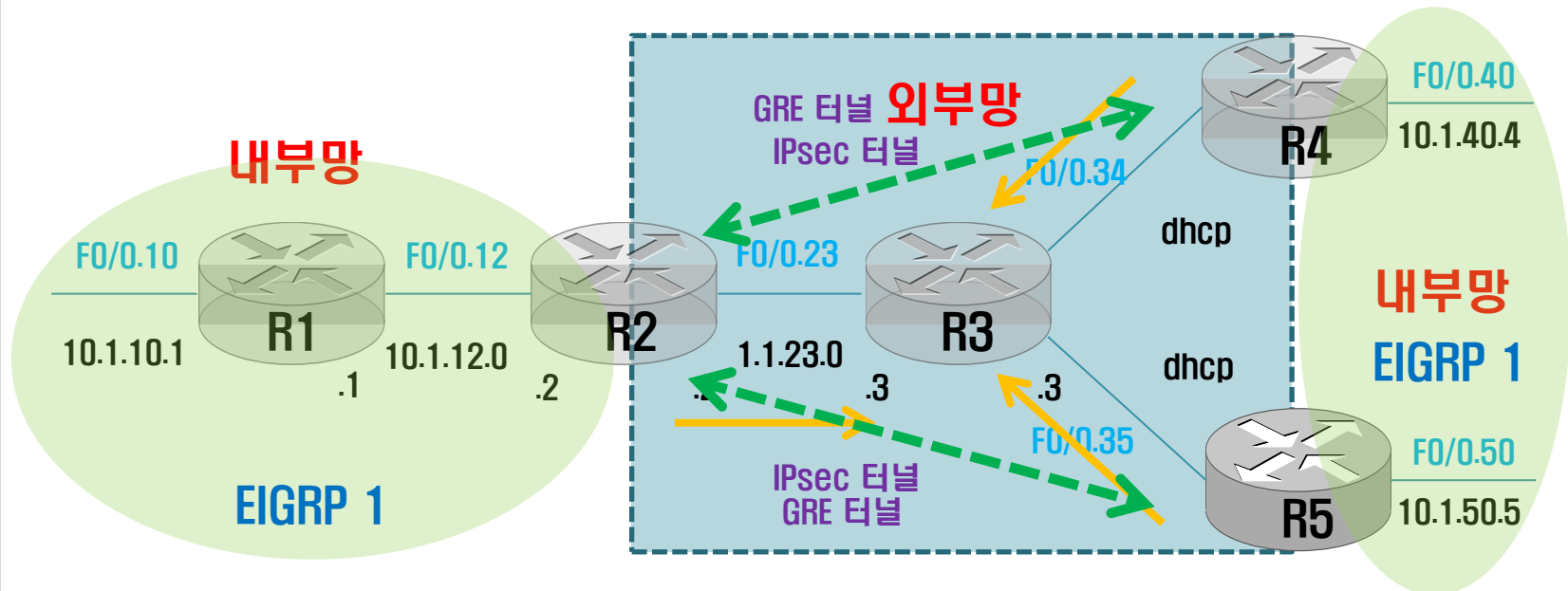
R2#show ip access-lists

(생략)

Extended IP access list acl_inbound

10 permit udp host 1.1.34.4 host 1.1.23.2 eq isakmp (15 matches)
20 permit udp host 1.1.35.5 host 1.1.23.2 eq isakmp (15 matches)
30 permit esp host 1.1.34.4 host 1.1.23.2 (66 matches)
40 permit esp host 1.1.35.5 host 1.1.23.2 (66 matches)
50 permit gre host 1.1.34.4 host 1.1.23.2
60 permit gre host 1.1.35.5 host 1.1.23.2

• 동적 크립토 맵 GRE IPsec VPN 구성



//추가해야 하는 것

<R3>

```
ip route 10.1.2.2 255.255.255.255 1.1.23.2
ip route 10.1.4.4 255.255.255.255 FastEthernet 0/0.34
ip route 10.1.5.5 255.255.255.255 FastEthernet 0/0.35
```

• 동적 크립토 맵 GRE IPsec VPN 구성

//GRE 터널 구성

//내부망 IP 주소를 터널 출발지/목적지 주소로 사용할 경우 네트워크를 라우팅에 포함시켜야 하고, 터널 목적지 주소를 터널을 통하여 광고 받아 터널이 다운 됨. 터널을 구성하는 한 쪽이 유동 IP주소를 사용하므로 라우팅 불필요한 별도 루프백 인터페이스에 IP 주소를 터널 출발지/목적지 주소로 사용.

<R2>

```
interface loopback 0
ip address 10.1.2.2 255.255.255.255
!
interface tunnel 24
ip address 10.1.24.2 255.255.255.0
tunnel source loopback 0
tunnel destination 10.1.4.4
keepalive 10 3
!
interface tunnel 25
ip address 10.1.25.2 255.255.255.0
tunnel source loopback 0
tunnel destination 10.1.5.5
keepalive 10 3
!
```

//지사 라우터 GRE 터널 설정
<R4>

```
interface loopback 0
ip address 10.1.4.4 255.255.255.255
!
interface tunnel 24
ip address 10.1.24.4 255.255.255.0
tunnel source loopback 0
tunnel destination 10.1.2.2
keepalive 10 3
!
```

<R5>

```
interface loopback 0
ip address 10.1.5.5 255.255.255.255
!
interface tunnel 25
ip address 10.1.25.5 255.255.255.0
tunnel source loopback 0
tunnel destination 10.1.2.2
keepalive 10 3
!
```

• 동적 크립토 맵 GRE IPsec VPN 구성

//GRE를 통한 동적 라우팅 설정
//터널 출발지 IP 주소는 라우팅 설정시 반드시 제외
//R2에서 10.1.2.2 네트워크를 EIGRP에 포함시키면 R4, R5에서 IP 목적지 주소인 10.1.2.2에 대한 라우팅 광고를 터널을 통하여 수신, 터널의 목적지 경로가 터널을 통하는 '재귀적 라우팅 (recursive routing)', 터널이 down 됨

<R1>

```
router eigrp 1
network 10.1.10.1 0.0.0.0
network 10.1.12.1 0.0.0.0
!
```

<R2>

```
router eigrp 1
network 10.1.12.2 0.0.0.0
network 10.1.24.2 0.0.0.0
network 10.1.25.2 0.0.0.0
!
```

<R4>

```
router eigrp 1
network 10.1.40.4 0.0.0.0
network 10.1.24.4 0.0.0.0
!
```

<R5>

```
router eigrp 1
network 10.1.50.5 0.0.0.0
network 10.1.25.5 0.0.0.0
!
```


• 동적 크립토 맵 GRE IPsec VPN 구성

//GRE IPsec VPN 구성

<R2>

```
crypto isakmp policy 10
encryption aes
authentication pre-share
```

!

//PSK 설정시 피어가 유동 IP 주소를 사용하므로 불특정 IP 주소를 의미하는 0.0.0.0 0.0.0.0

```
crypto isakmp key cisco address 0.0.0.0 0.0.0.0
```

!

```
crypto isakmp keepalive 10
```

!

```
ip access-list extended acl_GRE
permit gre host 10.1.2.2 host 10.1.4.4
permit gre host 10.1.2.2 host 10.1.5.5
```

!

```
crypto ipsec transform-set trset_GRE esp-aes
esp-md5-hmac
```

!

//피어가 유동 IP 주소 사용하므로 동적 크립토 맵 사용하여 현재 시점에서 결정된 IPSec SA만 지정

```
crypto dynamic-map dmap_GRE 10
match address acl_GRE
set transform-set trset_GRE
```

!

//크립토 맵에서 앞서 설정한 동적 크립토 맵 참조

```
crypto map cmap_GRE 10 ipsec-isakmp
dynamic dmap_GRE
```

!

```
interface f0/0.23
```

```
crypto map cmap_GRE
```

!

<R4>

```
crypto isakmp policy 10
encryption aes
authentication pre-share
```

!

```
crypto isakmp key cisco address 1.1.23.2
```

!

```
crypto isakmp keepalive 10
```

!

```
ip access-list extended acl_GRE
permit gre host 10.1.4.4 host 10.1.2.2
```

!

```
crypto ipsec transform-set trset_GRE esp-aes
esp-md5-hmac
```

!

• 동적 크립토 맵 GRE IPsec VPN 구성

<R4-이어서>

```
crypto map cmap_GRE 10 ipsec-isakmp
match address acl_GRE
set peer 1.1.23.2
set transform-set trset_GRE
!
interface f0/0.34
crypto map cmap_GRE
!
```

<R5>

```
crypto isakmp policy 10
encryption aes
authentication pre-share
!
crypto isakmp key cisco address 1.1.23.2
!
crypto isakmp keepalive 10
!
ip access-list extended acl_GRE
permit gre host 10.1.5.5 host 10.1.2.2
!
```

```
crypto ipsec transform-set trset_GRE esp-aes
esp-md5-hmac
```

!

```
crypto map cmap_GRE 10 ipsec-isakmp
match address acl_GRE
set peer 1.1.23.2
set transform-set trset_GRE
```

!

```
interface f0/0.35
crypto map cmap_GRE
!
```

• 동적 크립토 맵 GRE IPsec VPN 구성

//확인

R5#show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

NI - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route

Gateway of last resort is 1.1.35.3 to network 0.0.0.0

1.0.0.0/24 is subnetted, 1 subnets

C 1.1.35.0 is directly connected, FastEthernet0/0.35

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks

D 10.1.10.0/24 [90/297249536] via 10.1.25.2, 00:00:08, Tunnel25

D 10.1.12.0/24 [90/297246976] via 10.1.25.2, 00:00:08, Tunnel25

C 10.1.5.5/32 is directly connected, Loopback0

C 10.1.25.0/24 is directly connected, Tunnel25

D 10.1.24.0/24 [90/310044416] via 10.1.25.2, 00:00:08, Tunnel25

D 10.1.40.0/24 [90/310046976] via 10.1.25.2, 00:00:08, Tunnel25

C 10.1.50.0/24 is directly connected, FastEthernet0/0.50

S* 0.0.0.0/0 [254/0] via 1.1.35.3

• 동적 크립토 맵 GRE IPsec VPN 구성

//확인

R5#ping 10.1.40.4

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.1.40.4, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 180/226/256 ms

R2#show crypto map

Crypto Map "cmap_GRE" 10 ipsec-isakmp

Dynamic map template tag: dmap_GRE

Crypto Map "cmap_GRE" 65536 ipsec-isakmp

Peer = 1.1.34.1

Extended IP access list

access-list permit gre host 10.1.2.2 host 10.1.4.4

dynamic (created from dynamic map dmap_GRE/10)

Current peer: 1.1.34.1

Security association lifetime: 4608000 kilobytes/3600 seconds

PFS (Y/N): N

Transform sets={

trset_GRE,

}

• 동적 크립토 맵 GRE IPsec VPN 구성

//확인(이어서)

○ Crypto Map "cmap_GRE" 65537 ipsec-isakmp

Peer = 1.1.35.1

Extended IP access list

access-list permit gre host 10.1.2.2 host 10.1.5.5

dynamic (created from dynamic map dmap_GRE/10)

Current peer: 1.1.35.1

Security association lifetime: 4608000 kilobytes/3600 seconds

PFS (Y/N): N

Transform sets={

trset_GRE,

}

Interfaces using crypto map cmap_GRE:

FastEthernet0/0.23

R2#show crypto engine connections active

Crypto Engine Connections

ID	Interface	Type	Algorithm	Encrypt	Decrypt	IP-Address
1	Fa0/0.23	IPsec	AES+MD5	0	299	1.1.23.2
2	Fa0/0.23	IPsec	AES+MD5	301	0	1.1.23.2
3	Fa0/0.23	IPsec	AES+MD5	0	237	1.1.23.2
4	Fa0/0.23	IPsec	AES+MD5	239	0	1.1.23.2
1001	Fa0/0.23	IKE	SHA+AES	0	0	1.1.23.2
1002	Fa0/0.23	IKE	SHA+AES	0	0	1.1.23.2